

India's EV transition runs on two and three wheels — and the biggest opportunity is the undecided middle

Rose Mary Varghese · MS Data Science & Management, IIM Indore + IIT Indore · Based on a 2025 consumer survey and public registration, infrastructure and IEA data

Bottom line. India's EV market is real and still growing, but growth is slowing. It is dominated by two- and three-wheelers, not cars. Among surveyed consumers, more people are *undecided* about an EV than firmly committed either way. Their hesitation centres on three issues: charging access, purchase price and battery replacement cost. The cost case for an electric car depends mostly on how far the owner drives.

Implication. The most useful levers are the ones that resolve uncertainty: clear cost-of-ownership information for each customer's usage, guarantees on battery cost, and practical home charging. Demographic targeting is not supported by this evidence.

1.53M

EVs registered in India in 2023, up from ~2,400 in 2014

94%

of 2023 EV registrations were two- or three-wheelers

38%

of surveyed residents are undecided about an EV as their next vehicle

~5 yrs

breakeven for an electric car at a ₹4 lakh premium and 12,000 km a year

Key findings

- Growth is strong but slowing.** Annual EV registrations tripled in 2022, grew 49% in 2023 and about 27% in 2024 (roughly 1.95 million vehicles). The market is moving out of its early, explosive phase. Simple trend projections either badly under- or over-shot 2024, so plans should not rely on any single growth forecast.
- India's EV story is a two- and three-wheeler story.** Two-wheelers were 56% of 2023 EV registrations and three-wheelers 38%; cars were 5%. In passenger cars alone, India is far behind: about 2% of new car sales were electric in 2023, compared with 18% globally and 38% in China. Measured on cars only, India looks like a laggard; measured on all vehicles, the picture is very different.
- Adoption is concentrated, and it tracks charging, but cause and effect are unclear.** Ten states account for 79% of registrations, with Uttar Pradesh alone at 18%. States with more public chargers have more EVs, but bigger markets have more of both, and chargers follow demand as much as they create it. Notably, Uttar Pradesh, Bihar and Assam have far more EVs than their charger counts would suggest. These are e-rickshaw markets, where vehicles charge at depots or at home.
- Consumers are open but uncommitted.** Among surveyed Indian residents, 44% say their next vehicle is likely to be an EV, 38% are unsure, and 18% say it is unlikely. The undecided group is the largest single block.
- Three concerns dominate, and they are roughly equal.** Charging infrastructure (56%), high upfront cost (51%) and battery replacement cost (50%) are each cited by about half of respondents. The survey cannot rank one above the others. Safety (27%) and performance (17%) matter much less.
- People want to charge at home.** 64% prefer home charging, against 37% for public stations.
- Who people are does not predict intent; what they believe does.** Age, income, education, location type and vehicle ownership did not distinguish likely buyers from others. Overall opinion of EVs did, but that largely restates intent.
- Whether an electric car saves money depends on usage.** With a ₹4 lakh price premium, an electric car recovers its extra cost in about 3 years for a heavy user (20,000 km a year), about 5 years for a typical user (12,000 km) and about 10 years for a light user (6,000 km).

What this means

The market is won or lost in the undecided middle.

Roughly four in ten consumers have not made up their minds. Their stated concerns (charging, price, battery cost) are practical rather than ideological. Practical concerns can be answered with better information, products and guarantees.

"Is an EV cheaper?" has no single answer. The honest answer is "it depends on how much you drive." A generic savings claim will be wrong for many buyers and, for light users, could mislead.

Charging strategy should follow the segment. Two-wheeler and car buyers say they want home charging. Three-wheeler markets have grown with little public infrastructure. Counting public chargers is a weak proxy for the charging that actually enables adoption.

Segmenting by demographics is not supported.

Nothing in this data suggests targeting by age, income or education. Usage (distance driven) and attitudes are the more promising axes, and both are observable.

Recommendations

- 1 Lead with personalised cost of ownership, and target high-mileage users first.** Give buyers a breakeven estimate based on their own distance and the actual price gap, rather than a generic savings figure. Prioritise segments where the case is strongest and quickest: commuters with long daily distances, delivery and ride-hailing fleets, and commercial three-wheeler operators.
- 2 Take battery-cost risk off the buyer.** Half of respondents worry about battery replacement cost. Extended battery warranties, guaranteed buy-back values or battery subscriptions address this directly. These should be piloted and measured, because the survey shows the concern exists but not which remedy works.
- 3 Make home and workplace charging easy before scaling public networks.** Support for installing chargers in apartment complexes, help with electricity connections, and bundled home-charger offers match where consumers say they want to charge. Public investment is best aimed at highways and dense urban areas where home charging is not possible.
- 4 Plan by state type, not one national strategy.** E-rickshaw-led states (e.g. Uttar Pradesh, Bihar, Assam) need depot charging, financing and battery-swapping for operators. Two-wheeler and car-led states (e.g. Karnataka, Kerala, Maharashtra) need consumer-facing home charging and cost transparency.
- 5 Validate before committing major spend.** Confirm the barrier shares and the size of the undecided group with a larger, representative survey (400+ respondents, set quotas by state), and track actual purchases rather than stated intent.

Limitations

- The survey is not representative of India.** It reached 204 people through personal networks, over-represents Kerala, women and graduates, and includes many Indians living abroad. The headline consumer findings use only the 125 respondents living in India, so every percentage carries an uncertainty of roughly ± 9 points.
- Intent is not behaviour.** People who say they will buy an EV often do not. The 44% figure is an upper bound on near-term buying, not a forecast.
- Associations, not causes.** No finding here shows that one factor *causes* adoption, including the link between chargers and EV sales.
- Cost figures are illustrative.** The breakeven analysis uses average prices and running costs. It excludes resale value, insurance, financing and battery replacement, any of which can shift the answer.
- Data gaps.** National registration figures exclude Telangana. Charging-station data has no recorded date. International comparisons cover cars only.

Sources: own consumer survey (June 2025, n = 204; 125 Indian residents); Vahan-derived state registration data 2014–2023; Ministry of Power charging-station data; Rajya Sabha answers (Sessions 259 and 265); IEA Global EV Outlook 2024; 2024 registrations as reported from Vahan by the Ministry of Heavy Industries. Full method and code: github.com/malu3110/EV-Project.